



COMPETENCY STANDARD

FOR

ELECTRICAL TERMINATOR AND FITTER

(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Electrical Terminator and Fitter is a comprehensive document designed to assist in developing curricula, teaching materials, learning resources, and assessment tools. It serves as a framework for delivering industry-relevant training, ensuring that individuals who meet the outlined standards through assessment are well-qualified and prepared for employment.

This document has been specifically developed for training purposes under **the Skills for Industry Competitiveness and Innovation Program (SICIP)**, aiming to address the skills requirements of the construction sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

Public and private institutes are encouraged to utilize the information provided in this competency standard for training activities that contribute to the development of Bangladesh's workforce.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:

Occupation Specific Competencies (120 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-ETF-01-O	Understand electrical system	1. Interpret fundamental principles of electricity 2. Identify electrical circuit 3. Identify wires and cables 4. Identify fittings used in electrical works 5. Measure current and voltage	05
SICIP-CON-ETF-02-O	Perform Installation of electrical circuit	1. Prepare for works 2. Control one lamp by SPST and SPDT switch 3. Control one calling bell from two point 4. Prepare the circuit of fluorescent lamp 5. Install ceiling fan with regulator	10
SICIP-CON-ETF-03-O	Perform wire or cable termination	1. Collect tools, equipment and materials 2. Identify different types of cables lugs 3. Connect cables (1-4 mm²) 4. Maintain workplace, tools and materials	10
SICIP-CON-ETF-04-O	Perform cable glanding	1. Collect tools, equipment and materials 2. Identify different type of cables gland 3. Connect PG-gland 4. Maintain workplace, tools and materials	15
SICIP-CON-ETF-05-O	Perform trunking work	1. Collect tools, equipment and materials 2. Perform PVC trunking wiring 3. Install boards and other accessories of wiring 4. Test wiring 5. Maintain workplace, tools and materials	40
SICIP-CON-ETF-06-O	Perform pipe wiring	1. Collect tools, equipment and materials 2. Install PVC pipe and set cables	40

Code	Unit of Competency	Elements of Competency	Duration (hours)
		3. Install boards and other accessories of wiring 4. Test wiring 5. Maintain workplace, tools and materials	
Total Hours			120

COMPETENCY STANDARD: ELECTRICAL TERMINATOR AND FITTER

A. Occupation-specific Competencies

Unit of Competency: UNDERSTAND ELECTRICAL SYSTEM	Nominal Duration: 05 Hrs.	Unit Code: SICIP-CON-ETF-01-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to understand electrical system. It specifically includes the tasks of interpreting fundamental principles of electricity, identifying electrical circuit, identifying wires and cables, identifying fittings used in electrical works and measuring current and voltage.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret fundamental principles of electrical	1.1 <u>Fundamental principles</u> of electrical are described. 1.2 Application of Ohm's law is explained. 1.3 <u>Electrical properties/parameters</u> are comprehended
2. Identify electrical circuit	2.1 <u>Basic elements</u> for preparing electrical circuits are identified and described. 2.2 <u>Electrical circuit diagram</u> is understood and explained. 2.3 <u>Electrical circuit</u> wirings are described.
3. Identify wires and cables	3.1 Different <u>types of wires</u> are identified and explained. 3.2 Different <u>types of cables</u> are identified and described. 3.3 <u>Size of cables</u> is interpreted according to specifications and standards. 3.4 <u>Capacity of cables</u> is understood including their electrical and mechanical limits.
4. Identify fittings used in electrical works	4.1 <u>Various types of electrical fittings</u> are identified according to the specifications and work requirements. 4.2 Specifications and characteristics of fittings are interpreted from technical documents and manuals. 4.3 Compatibility of fittings with cables, conduits, and electrical devices is identified.
5. Measure current and voltage	5.1 Appropriate <u>measuring instruments</u> are identified and selected according to the measurement requirements. 5.2 Safety procedures are applied before and during measurement. 5.3 Voltage is measured across the designated points. 5.4 Current is measured in the circuit by connecting the instruments.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Fundamental principles	1.1 Ohm's Law 1.2 Principles and theory of AC/DC circuits 1.3 Series and parallel circuits 1.4 Semiconductor 1.5 Half wave and full wave rectification
2. Electrical properties/parameters	2.1 Voltage 2.2 Current 2.3 Resistance 2.4 Capacitance 2.5 Inductance
3. Basic elements	3.1 Switch 3.2 Socket 3.3 Cable 3.4 Resistive load 3.5 Inductive load
4. Electrical circuit diagram	4.1 Series circuit 4.2 Parallel circuit 4.3 Series-parallel circuit 4.4 Lamp circuits
5. Electrical circuit	5.1 Series circuits 5.2 Parallel circuits 5.3 Series-parallel circuits 5.4 One Lamp controlled from one point 5.5 One Lamp controlled from two points 5.6 One Lamp controlled from three points 5.7 Two Lamps controlled from two points 5.8 Two Lamps controlled from three points 5.9 Tube Light Connection 5.10 Power supply circuit
6. Types of wire	6.1 Solid Wire 6.2 Single-Conductor Wire 6.3 Stranded Wire 6.4 Multi-Conductor Wire
7. Types of cables	7.1 Low voltage 7.2 Medium voltage 7.3 High voltage 7.4 Flexible cable 7.5 Power cable 7.6 Armored cable 7.7 Fire resistance cable 7.8 Instrumentation cable
8. Size of cables	8.1 1.5mm ² 8.2 2.5mm ²

	8.3 4mm ²
9. Capacity of cables	9.1 2.5mm ² copper cable typically has a current-carrying capacity around 28A. 9.2 4mm ² copper cable can typically handle up to 35A.
10. Various types of electrical fittings	10.1 Cable Glands 10.2 Conduits and Conduit Fittings 10.3 Junction Boxes 10.4 Switches and Switch Boxes 10.5 Socket Outlets and Plug Sockets 10.6 Clips and Clamps 10.7 Terminals and Terminal Blocks 10.8 Cable Trunking and Trunking Accessories 10.9 Lighting Fittings
11. Measuring instruments	11.1 Ammeter 11.2 Clamp Meter 11.3 Voltmeter 11.4 Multimeter

Curricular Content Guide

1. Underpinning Knowledge	1.1 Describe fundamental principles of electricity and their applications. 1.2 Explain the application of Ohm's Law in electrical circuits and calculations. 1.3 Comprehend key electrical properties and parameters. 1.4 Identify and describe the basic elements. 1.5 Understand and explain electrical circuit diagrams, symbols, and layouts. 1.6 Describe methods and techniques used for electrical circuit wiring. 1.7 Identify different types of wires, their characteristics, and appropriate uses. 1.8 Identify various types of cables and describe their construction, applications, and specifications. 1.9 Interpret cable sizes according to national/international specifications and standards. 1.10 Understand the electrical and mechanical capacity limits of cables to ensure safe and efficient operation. 1.11 Identify various types of electrical fittings based on their specifications and work requirements.
2. Underpinning Skills	2.1 Applying Ohm's law in preparing a circuit 2.2 Preparing series and parallel electrical circuits 2.3 Wiring electrical circuits 2.4 Using different types of wires 2.5 Using different types of cables

	2.6 Measuring cable sizes according to specifications and standards 2.7 Measuring the capacity of cables, including their electrical and mechanical limits 2.8 Using various types of electrical fittings according to specifications and work requirements 2.9 Selecting appropriate measuring instruments according to measurement requirements 2.10 Applying safety procedures before and during measurement 2.11 Measuring voltage across the designated points 2.12 Measuring current in the circuit by connecting the instruments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Various kinds of electrical components, tools and equipment 4.3 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted fundamental principles of electrical 1.2 identified electrical circuit 1.3 identified wires and cables 1.4 identified fittings used in electrical works 1.5 measured current and voltage
2. Methods of Assessment	2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM INSTALLATION OF ELECTRICAL CIRCUIT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-CON-ETF-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform installation of electrical circuit. It specifically includes the tasks of preparing for works, controlling one lamp by SPST and SPDT switch, controlling one calling bell from two point, preparing the circuit of fluorescent lamp and installing ceiling fan with regulator.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for works	1.1 <u>Personal protective equipment (PPE)</u> is identified and used. 1.2 Work area is inspected for hazards and prepared to ensure a safe environment. 1.3 Relevant work instructions, drawings, and specifications are understood. 1.4 Required materials, fittings, cables, and tools are arranged according to the job requirements.
2. Control one lamp by SPST and SPDT switch	2.1 Circuit diagrams, SPST(Single pole single through) and SPDT(Single Pole Double Through) switch connections are reviewed and understood. 2.2 SPST switch, SPDT switch, lamp, wiring, and other necessary materials are selected. 2.3 Wires are measured, cut, stripped, and connected to the SPST and SPDT switches and lamp terminals. 2.4 Electrical connections are securely tightened and properly insulated to prevent faults. 2.5 Lamp is controlled and tested by operating the SPST and SPDT switches to ensure correct ON/OFF operation.
3. Control one calling bell from two point	3.1 Instructions and wiring diagrams related to the calling bell control circuit are reviewed and understood. 3.2 Calling bell unit, wiring cables, switches, and necessary tools are identified and prepared. 3.3 Wiring is measured, cut, and stripped according to the circuit layout. 3.4 Wires are connected securely to the calling bell and the controlling switch terminals. 3.5 The calling bell is operated from the control point to verify that it rings correctly.
4. Prepare the circuit of fluorescent lamp	4.1 Fluorescent lamp circuit diagram and technical specifications are reviewed and understood.

	4.2 All necessary components are identified and arranged. 4.3 Wires are measured, cut, and stripped 4.4 Wiring connections between lamp holder, switch, and power supply are made. 4.5 Terminals and joints are securely tightened and insulated to ensure safety and proper functioning. 4.6 Fluorescent lamp circuit is tested to verify that the lamp lights up correctly when switched ON.
5. Install ceiling fan with regulator	5.1 Relevant installation instructions and wiring diagrams for the ceiling fan and regulator are reviewed and understood. 5.2 Ceiling fan, regulator, mounting accessories, wiring, and necessary tools are identified and arranged. 5.3 Wiring is measured according to specifications, cut to length, stripped, and prepared for connection. 5.4 Fan is mounted on the ceiling and the regulator is installed at the designated location. 5.5 Electrical Connections are made as per wiring diagram connections. 5.6 Ceiling fan and regulator are tested to confirm correct speed control and safe operation.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety helmet 1.4 Hand gloves 1.5 Safety goggles 1.6 Safety shoes
2. All necessary components	2.1 Fluorescent tube 2.2 Switch 2.3 Wiring 2.4 Holders

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand relevant work instructions, electrical drawings, and technical specifications. 1.2 Review and understand circuit diagrams. 1.3 SPST switches, SPDT switches, lamps, wiring, and other necessary materials. 1.4 Measure, cut, strip, and connect wires to switches and lamp terminals. 1.5 Review instructions and wiring diagrams related to calling
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	bell control circuits. 1.6 Identify calling bell units, wiring cables, switches, and necessary tools. 1.7 Review fluorescent lamp circuit diagrams and technical specifications.
2. Underpinning Skills	2.1 Inspecting the work area for hazards and preparing a safe environment 2.2 Arranging required materials, fittings, cables, and tools according to job requirements 2.3 Measuring, cutting, stripping, and connecting wires to SPST and SPDT switches and lamp terminals 2.4 Operating and testing the SPST and SPDT switches to control and verify lamp operation 2.5 Connecting wires securely to the calling bell and control switch terminals 2.6 Operating the calling bell from the control point to verify correct operation 2.7 Testing the fluorescent lamp circuit to verify the lamp lights up correctly when switched ON 2.8 Mounting the ceiling fan and installing the regulator at the designated location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools and equipment appropriate to joint and connection process 4.3 Availability of materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for works 1.2 controlled one lamp by SPST and SPDT switch 1.3 controlled one calling bell from two point 1.4 prepared the circuit of fluorescent lamp 1.5 installed ceiling fan with regulator
2. Methods of Assessment	2.1 Written Test

	2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM WIRE OR CABLE TERMINATION	Nominal Duration: 10 Hrs.	Unit Code: SICIP-CON-ETF-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform wire or cable termination. It specifically includes the tasks of collecting tools, equipment and materials, identifying different types of cables lugs, connecting cables (1-4 mm ²) and maintaining workplace, tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials	1.1 <u>Personal protective equipment (PPE)</u> is identified and used. 1.2 <u>Appropriate tools and equipment</u> are identified and collected based on the termination requirements. 1.3 <u>Materials</u> of termination works are identified and arranged as per termination requirements.
2. Identify different types of cables lugs	2.1 Specifications, datasheets, and manufacturer manuals related to cable lugs are reviewed. 2.2 <u>Different types of cable lugs</u> are identified. 2.3 <u>Cable lug sizes</u> are identified to ensure compatibility with cables and electrical requirements. 2.4 Compatibility of cable lugs and sizes is confirmed based on technical data.
3. Connect cables (1-4 mm ²)	3.1 Cables sized 1-4 mm ² are measured, cut to length, and insulation is stripped. 3.2 Cable ends are cleaned, and conductors are twisted or tinned if required. 3.3 Appropriate connectors or terminals compatible with 1-4 mm ² cables are selected. 3.4 Cables are connected to terminals or connectors using crimping. 3.5 All connections are insulated with heat shrink tubing, or electrical tape. 3.6 Electrical continuity is tested using appropriate testing instruments.
4. Maintain workplace, tools and materials	4.1 Workplace is cleaned and organized to ensure safety and efficiency. 4.2 Tools are inspected regularly for damage or wear and maintained according to manufacturer guidelines.

	4.3 Tools and equipment are stored in designated areas following safety and organizational procedures.
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Range of Variables

Variable	Range (Includes but not limited to:)
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety helmet 1.4 Hand gloves 1.5 Safety goggles 1.6 Safety shoes
2. Appropriate tools and equipment	2.1 Wire cutter 2.2 Wire stripper 2.3 Knife 2.4 Crimping tools
3. Materials	3.1 Cables 3.2 Connectors/lugs 3.3 Terminals 3.4 Insulating materials
4. Different types of cable lugs	4.1 Ring/O Lugs 4.2 Fork/U Lugs 4.3 Pin/I Lugs
5. Cable lug sizes	5.1 1.5mm ² 5.2 2.5mm ² 5.3 4mm ²

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment required for cable termination tasks. 1.2 Review and interpret specifications, datasheets, and manufacturer manuals. 1.3 Identify different types of cable lugs used in electrical terminations and their specific applications. 1.4 Identify cable lug sizes and assess their compatibility with cables and electrical requirements. 1.5 Prepare cable ends by cleaning, twisting, or tinning conductors. 1.6 Inspect tools regularly for damage or wear.
2. Underpinning Skills	2.1 Collecting appropriate tools and equipment based on termination requirements 2.2 Arranging materials for termination works according to requirements

	2.3 Measuring, cutting, and stripping insulation from cables sized 1-4 mm ² 2.4 Connecting cables to terminals or connectors using crimping 2.5 Insulating all connections with heat shrink tubing or electrical tape 2.6 Testing electrical continuity using appropriate testing instruments 2.7 Inspecting tools regularly for damage or wear and maintaining them according to manufacturer guidelines 2.8 Storing tools and equipment in designated areas following safety and organizational procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools and equipment required for the job 4.3 Materials required 4.4 Work books 4.5 Diagram and instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected tools, equipment and materials 1.2 identified different types of cables lugs 1.3 connected cables (1-4 mm ²) 1.4 maintained workplace, tools and materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM CABLE GLANDING	Nominal Duration: 15 Hrs.	Unit Code: SICIP-CON-ETF-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform cable glanding. It specially includes the tasks of collecting tools, equipment and materials, identifying different types of cables gland, connecting PG-gland and maintaining workplace tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials	1.1 <u>Personal protective equipment (PPE)</u> is identified and used. 1.2 <u>Appropriate tools and equipment</u> are identified and collected based on the glanding requirements. 1.3 <u>Materials</u> of glanding works are identified and arranged as per glanding requirements.
2. Identify different type of cables gland	2.1 Technical manuals, datasheets, and relevant standards are understood. 2.2 <u>Different types of cable glands</u> are identified. 2.3 Materials used in cable glands and their specific applications are understood. 2.4 Cable gland sizes and thread types are identified against cable and conduit specifications.
3. Connect PG-gland	3.1 Relevant PG-gland specifications and installation instructions are understood. 3.2 Appropriate PG-gland size is selected based on the cable diameter and type. 3.3 Cable is measured, cut, and stripped as per instruction sheet. 3.4 <u>PG-gland components</u> are assembled on the cable in correct order. 3.5 Cable is inserted through the gland ensuring a comfortable and secure fit of the sealing ring. 3.6 PG-gland is tightened using appropriate tools.
4. Maintain workplace tools and materials	4.1 Workplace is cleaned and organized to ensure safety and efficiency. 4.2 Tools are inspected regularly for damage or wear and maintained according to manufacturer guidelines. 4.3 Tools and equipment are stored in designated areas following safety and organizational procedures.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety helmet 1.4 Hand gloves 1.5 Safety goggles 1.6 Safety shoes
2. Appropriate tools and equipment	2.1 Cutters 2.2 Knife 2.3 Wrench/Spanner 2.4 Measuring tape
3. Materials	3.1 Cables 3.2 Glands 3.3 Insulating materials 3.4 Marker pen
4. Different types of cable glands	4.1 Metallic 4.2 Non-metallic 4.3 PG-gland 4.4 Explosion-proof 4.5 Armored 4.6 Unarmored
5. PG-gland components	5.1 Main Body 5.2 Clamping Ring/Claw 5.3 Seal 5.4 Locknut/Sealing Nut

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment required for cable glanding tasks. 1.2 Recognize materials used in glanding works and arrange them according to job requirements. 1.3 Understand technical manuals, datasheets, and relevant industry standards related to cable glands. 1.4 Identify different types of cable glands and their specific applications. 1.5 Understand materials used in cable glands and their appropriate uses in various conditions. 1.6 Identify cable gland sizes and thread types according to cable and conduit specifications. 1.7 Understand relevant PG-gland specifications and installation instructions. 1.8 Understand PG-gland size based on cable diameter and type.
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	1.9 Insert cable through the gland ensuring a secure fit of the sealing ring.
2. Underpinning Skills	2.1 Collecting appropriate tools and equipment based on glanding requirements 2.2 Arranging materials for glanding works according to requirements 2.3 Measuring, cutting, and stripping cables as per instruction sheet 2.4 Assembling PG-gland components on the cable in the correct order 2.5 Inserting the cable through the gland ensuring a comfortable and secure fit of the sealing ring 2.6 Cleaning and organizing the workplace to ensure safety and efficiency
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Materials, tools and equipment needed for the activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected tools, equipment and materials 1.2 identified different type of cables gland 1.3 connected PG-gland 1.4 maintained workplace tools and materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM TRUNKING WORK	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-ETF-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform trunking work. It specially includes the tasks of collecting tools, equipment and materials, performing PVC trunking wiring, installing boards and other accessories of wiring, testing wiring and maintaining workplace, tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials	1.1 <u>Personal protective equipment (PPE)</u> is identified and used. 1.2 <u>Appropriate tools and equipment</u> are identified and collected based on the trunking requirements. 1.3 <u>Materials</u> of termination works are identified and arranged as per trunking requirements.
2. Perform PVC trunking wiring	2.1 Relevant work instructions, wiring diagrams, and trunking layout plans are reviewed and understood. 2.2 PVC trunking, cables, connectors, fixing accessories are identified and arranged. 2.3 Route for PVC trunking installation is measured and marked on surfaces according to layout plans. 2.4 PVC trunking is cut to required lengths and edges are smoothed. 2.5 PVC trunking sections are fixed firmly to walls. 2.6 Electrical cables are laid inside the trunking without damage. 2.7 Covers are placed and secured on the trunking to protect the cables.
3. Install boards and other accessories of wiring	3.1 Relevant installation instructions, wiring diagrams, and layout plans for boards and accessories are understood. 3.2 Boards switches, sockets, conduits are identified and prepared. 3.3 Boards and accessories are fixed firmly to walls. 3.4 Wiring is connected to the installed boards and accessories as per the circuit diagrams. 3.5 All connections are tightened, checked and insulated.
4. Test wiring	4.1 Appropriate <u>testing instruments</u> are selected for proper functioning. 4.2 Relevant work instructions, testing procedures, and safety protocols are understood before testing.

	4.3 Test points on wiring circuits are identified according to wiring diagrams. 4.4 Insulation resistance is measured between conductors and earth using an insulation tester. 4.5 Continuity of conductors and connections is tested with faults noted. 4.6 Polarity of wiring connections is tested. 4.7 All test results are recorded and compared against standard values.
5. Maintain workplace, tools and materials	5.1 Workplace is cleaned and organized to ensure safety and efficiency. 5.2 Tools are inspected regularly for damage or wear and maintained according to manufacturer guidelines. 5.3 Tools and equipment are stored in designated areas following safety and organizational procedures.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety helmet 1.4 Hand gloves 1.5 Safety goggles 1.6 Safety shoes
2. Appropriate tools and equipment	2.1 Cutters 2.2 Wrench/Spanner 2.3 Measuring tape 2.4 Hand grinder/hack saw 2.5 Screw-driver 2.6 Combination pliers 2.7 Neon tester 2.8 Cutting pliers 2.9 Center punch
3. Materials	3.1 PVC trunking 3.2 Screw 3.3 Washer 3.4 Marker pen 3.5 Water level 3.6 Try square 3.7 Insulation tape
4. Testing instruments	4.1 Insulation resistance tester/Megger 4.2 Multimeter/Ohm meter 4.3 Test lamp

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment required for PVC trunking installation tasks. 1.2 Identify PVC trunking, cables, connectors, and fixing accessories. 1.3 Cutting procedures of PVC trunking. 1.4 Understand installation procedures to electrical boards and accessories. 1.5 Identify boards, switches, sockets, and conduits for installation. 1.6 Identify appropriate testing instruments. 1.7 Understand work instructions, testing procedures. 1.8 Identify test points on wiring circuits.
2. Underpinning Skills	1.1 Collecting appropriate tools and equipment based on trunking requirements 1.2 Arranging materials for termination works according to trunking requirements 1.3 Measuring and marking the route for PVC trunking installation on surfaces according to layout plans 1.4 Cutting PVC trunking to required lengths and smoothing the edges 1.5 Fixing PVC trunking sections firmly to walls 1.6 Laying electrical cables inside the trunking without damage 1.7 Placing and securing covers on the trunking to protect cables 1.8 Fixing boards, switches, sockets, and conduits firmly to walls 1.9 Connecting wiring to installed boards and accessories according to circuit diagrams 1.10 Tightening, checking, and insulating all connections 1.11 Testing continuity of conductors and connections, noting faults 1.12 Testing polarity of wiring connections
3. Underpinning Attitudes	1.13 Commitment to occupational health and safety 1.14 Promptness in carrying out activities 1.15 Sincerity and honesty to duties 1.16 Environmental concerns 1.17 Eagerness to learn 1.18 Tidiness and timeliness 1.19 Respect for rights of peers and seniors in workplace 1.20 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided; 1.21 Workplace (actual or simulated)

	1.22 Materials relevant to the proposed activity 1.23 Measuring instrument 1.24 Relevant drawings, manuals, codes, standards and reference materials
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected tools, equipment and materials 1.2 performed PVC trunking wiring 1.3 installed boards and other accessories of wiring 1.4 tested wiring 1.5 maintained workplace, tools and materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM PIPE WIRING	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-ETF-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform pipe wiring. It specially includes the tasks of collecting tools, equipment and materials, installing PVC pipe and setting cables, installing boards and other accessories of wiring, testing wiring and maintaining workplace, tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials	1.1 <u>Personal protective equipment (PPE)</u> is identified and used. 1.2 <u>Appropriate tools and equipment</u> are identified and collected based on the pipe wiring requirements. 1.3 <u>Materials</u> of pipe wiring works are identified and arranged as per pipe wiring requirements.
2. Install PVC pipe and set cables	2.1 Relevant installation instructions and layout plans for PVC piping and cable routing are understood. 2.2 PVC pipes, fittings, cables, and required tools are identified and arranged. 2.3 Route for PVC pipe installation is measured and marked on surfaces according to the layout. 2.4 PVC pipes are cut to the required lengths and edges are smoothed to ensure proper joining. 2.5 Pipes and fittings are assembled and fixed to surface or concealed using appropriate clamps. 2.6 Electrical cables are carefully pulled through the installed PVC pipes without damage. 2.7 The PVC pipe installation and cable setting are inspected to ensure compliance with standards and specifications.
3. Install boards and other accessories of wiring	3.1 Wiring diagrams for boards and accessories are reviewed and understood. 3.2 Boards, switches, sockets, conduits, and other wiring accessories are identified and prepared. 3.3 Boards and accessories are securely fixed to walls, panels, using appropriate tools. 3.4 Wiring is connected to the installed boards and accessories. 3.5 All wiring connections are tightened and insulated to prevent electrical hazards and ensure reliability.
4. Test wiring	4.1 Appropriate <u>testing instruments</u> are selected for proper functioning.

	4.2 Relevant work instructions, testing procedures, and safety protocols are understood before testing. 4.3 Test points on wiring circuits are identified according to wiring diagrams. 4.4 Insulation resistance is measured between conductors and earth using an insulation tester. 4.5 Continuity of conductors and connections is tested with faults noted. 4.6 Polarity of wiring connections is tested. 4.7 All test results are recorded and compared against standard values.
5. Maintain workplace, tools and materials	5.1 Workplace is cleaned and organized to ensure safety and efficiency. 5.2 Tools are inspected regularly for damage or wear and maintained according to manufacturer guidelines. 5.3 Tools and equipment are stored in designated areas following safety and organizational procedures.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Personal protective equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety helmet 1.4 Hand gloves 1.5 Safety goggles 1.6 Safety shoes
2. Appropriate tools and equipment	2.1 Cutters 2.2 Measuring tape 2.3 Hack saw 2.4 Round file 2.5 Pipe bending spring 2.6 Flat screw-driver 2.7 Star screw-driver 2.8 Combination pliers 2.9 Neon tester 2.10 Cutting pliers 2.11 Center punch
3. Materials	3.1 PVC pipe 3.2 Clamps/saddle 3.3 Screw 3.4 Washer 3.5 Fish wire/cable pulling spring 3.6 Marker pen 3.7 Water level 3.8 Insulation tape

4. Testing instruments	4.1 Insulation resistance tester/Megger 4.2 Multimeter/Ohm meter 4.3 Test lamp
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand relevant installation instructions and layout plans for PVC piping and cable routing. 1.2 Identify PVC pipes, fittings, cables, and required tools. 1.3 PVC Pipe installation procedures. 1.4 PVC pipe cutting procedures. 1.5 Assembling process and pipe and fittings. 1.6 Inspecting procedures of PVC pipe installation and cable setting. 1.7 Connecting process of wiring to the installed boards and accessories. 1.8 Understand work instructions, testing procedures, and safety protocols before conducting tests.
2. Underpinning Skills	2.1 Collecting appropriate tools and equipment based on pipe wiring requirements 2.2 Arranging materials for pipe wiring works according to requirements 2.3 Measuring and marking the route for PVC pipe installation according to layout plans 2.4 Cutting PVC pipes to required lengths and smoothing the edges for proper joining 2.5 Assembling and fixing pipes and fittings to surfaces or concealing them using appropriate clamps 2.6 Pulling electrical cables through installed PVC pipes without damage 2.7 Inspecting PVC pipe installation and cable setting to ensure compliance with standards and specifications 2.8 Preparing boards, switches, sockets, conduits, and other wiring accessories 2.9 Fixing boards and accessories securely to walls or panels using appropriate tools 2.10 Measuring insulation resistance between conductors and earth using an insulation tester
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace

	3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided; 3.1 Workplace (actual or simulated) 3.2 Materials relevant to the proposed activity 3.3 Measuring instrument 3.4 Relevant drawings, manuals, codes, standards and reference materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected tools, equipment and materials 1.2 installed PVC pipe and set cables 1.3 installed boards and other accessories of wiring 1.4 tested the wiring 1.5 maintained workplace, tools and materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Workshop/Lab Facility Standard

Course Name:	Electrical Terminator and Fitter
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop with internet facility	01
2.	Multimedia projector & screen	01
3.	Wiring Board	20
4.	Main Distribution Board with busbar, CB and other accessories (Training model)	01
5.	Sub Distribution Board with busbar, CB and other accessories (Training model)	20
6.	Crimping tools	10
7.	Adjustable wrench	10
8.	Electric drill machine	02
9.	Electric grinding machine	02
10.	Energy Meter (1 Phase)	10
11.	Energy Meter (3 Phase)	05
12.	Multi meter/AVO meter	20
13.	Insulation tester	02
14.	Clamp Meter	02
15.	Phase sequence meter	01
16.	Ceiling fan	10
17.	Bench Vice	05
18.	Tool box with all accessories	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Electrical Terminator and Fitter, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.